

14. Shortest Seek Time First Disk Scheduling

```
#include <stdio.h>

int main () {
    int n, i, j;
    printf("Enter the number of requests: ");
    scanf("%d", &n);

    int req[n], visited[n];
    printf("Enter Requests: ");
    for (i=0;i<n;i++) {
        scanf("%d", &req[i]);
        visited[i]=0;
    }

    int head;
    printf("Enter the initial Head position: ");
    scanf("%d",&head);

    int total = 0;

    for (i=0;i<n;i++) {
        int min=100000, k=-1;

        for (j=0;j<n;j++) {
            if (!visited[j]){
                int diff = req[j] - head;
                if (diff<0) diff = -diff;

                if (diff < min){
                    min = diff;
                    k = j;
                }
            }
        }

        visited[k] = 1;
        total += min;
        head = req[k];
    }

    printf("Total seek time: %d\n", total);
    return 0;
}
```